고급통계분석 - Assignment02

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Problem 2.

(a).

```
gaussian_kernel_mat <- function(x, rho = 1){
  x <- as.numeric(x)
  outer(x, x, function(a, b) exp(-rho * (a - b)^2))
}
```

(b).

```
library(Rcpp)

cppFunction('
#include <Rcpp.h>
using namespace Rcpp;

// [[Rcpp::export]]
NumericMatrix gaussian_kernel_mat_cpp(const NumericVector& x, const
double rho = 1.0){
  int n = x.size();
  NumericMatrix K(n, n);
  for (int i = 0; i<n; ++i){
    K(i, i) = 1.0;
    for(int j = i+1; j<n; ++j){
      double d = x[i] - x[j];
      double d2 = d*d;
      double kij = std::exp(-rho * d2);
      K(i, j) = kij;
      K(j, i) = kij;
    }
  }
  return K;
}
')
```

```
## Warning: No function found for Rcpp::export attribute at
filed5f47ffe2706.cpp:5
```

(c).

```
set.seed(123)
n <- 1000
x <- runif(n)

times_r <- replicate(50, system.time(gaussian_kernel_mat(x,
1))[['elapsed']])
```

```r
times_rcpp <- replicate(50, system.time(gaussian_kernel_mat_cpp(x,
1))[['elapsed']])

mean(times_r); mean(times_rcpp)
```

```
## [1] 0.1022

## [1] 0.0474
```

(d).

```r
result <- data.frame(
  Method = c('R', 'Rcpp'),
  mean_time = c(mean(times_r), mean(times_rcpp))
)

print(result)
```

```
##   Method mean_time
## 1      R    0.1022
## 2   Rcpp    0.0474
```